

MPLADS IntelliTrack

AI-Powered Monitoring, Anomaly Detection & Decision-Support Platform

A comprehensive monitoring and decision-support platform for MPLADS implementation designed to digitally record operations, continuously analyze available data, detect unusual patterns, predict risks, and assist authorities in targeted human verification.

Core Architectural Principle: The system must not declare something as “fraud” merely because an anomaly is detected. It should identify an anomaly/risk, explain why it was flagged, and send it for human verification. That distinction is important for a government-grade system.

The proposed system is a monitoring and decision-support platform for MPLADS implementation. It does not replace the existing government approval, sanction, tender, payment, or physical-verification process. Instead, it digitally records what happens, continuously analyzes the available data, detects unusual patterns, predicts risks, and helps the concerned authorities decide where attention or verification is required.

1. Complete Services & Features

A. User, Access & Security Services

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
1	User Registration & Login	Authorized users can securely access the platform using government-issued credentials.
2	Role-Based Access Control (RBAC)	Different users see only the information relevant to their role.
3	MoSPI Dashboard	Provides a national-level view of MPLADS works, expenditure, risks, anomalies and trends across India.
4	State Authority Dashboard	Shows MPLADS information belonging to the respective state.
5	District Authority Dashboard	Provides detailed control and monitoring of works within the district.
6	MP Dashboard	Allows an MP to see works recommended under their constituency and their execution status.
7	Implementing Agency Access	Agencies can update project execution, tender and implementation information.
8	Contractor Access	Contractors can upload permitted project evidence such as progress photographs, bills and invoices.
9	Jurisdiction Control	A user cannot access projects outside their authorized geographic/ administrative jurisdiction.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
10	MFA / Secure Authentication	Adds an additional authentication layer for sensitive accounts.
11	Session Management	Controls active sessions and prevents unauthorized access.
12	Audit Login History	Records who accessed the system and when.

B. MPLADS Project Lifecycle Management

This module digitizes the monitoring record of the existing MPLADS process.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
13	MP Work Recommendation	MP/project office enters the recommended work and its basic details.
14	Work Description	Stores description, purpose, location, beneficiaries, estimated capacity and other project information.
15	Estimated Cost Entry	Records the estimated cost proposed for the work.
16	Project Location Registration	Stores the proposed geographical coordinates/location of the work.
17	District Scrutiny Record	District authority records scrutiny/verification information.
18	Sanction Information	Records sanction details after the competent authority approves the work.
19	Implementing Agency Assignment	Records which agency is responsible for execution.
20	Tender Information	Stores relevant tender/procurement information.
21	Contractor Assignment	Records the selected contractor against the project.
22	Unique Project ID	Every project receives a unique identifier for tracking throughout its lifecycle.
23	Project Status Tracking	Tracks stages such as proposed, sanctioned, tendered, ongoing, completed, etc.
24	Project Timeline	Stores planned start date, milestones and expected completion date.
25	Project Completion Record	Records completion information and asset creation details.

C. Data Collection & Data Entry

The system depends on accurate data from authorized stakeholders.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
26	Sanction Data Collection	Stores sanctioned amount, date, authority and related details.
27	Cost Estimate Collection	Stores estimated project cost and component-wise estimates.
28	Expenditure Data	Records actual expenditure against the project.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
29	Payment Data	Records payments made during execution.
30	Bill Upload	Contractors/agencies can submit bills through the system.
31	GST Invoice Upload	Required invoice information can be uploaded for verification.
32	Progress Data	Records percentage of physical work completed.
33	Milestone Updates	Individual project milestones can be updated and monitored.
34	Document Upload	Relevant project documents can be stored digitally.
35	Contractor Information	Stores contractor/project assignment information.
36	Implementing Agency Information	Records the agency responsible for execution.
37	Asset Information	Records the final asset created through the project.

D. Geospatial & GPS Monitoring

This is one of the stronger parts of your proposed solution.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
38	GPS Project Location	Stores the approved geographical location of the project.
39	Geo-Tagged Progress Photo	Progress photographs contain location information captured when the photograph is taken.
40	Timestamp Verification	Records when the evidence was captured.
41	Location Comparison	Compares the submitted evidence location with the registered project location.
42	GPS Mismatch Alert	Flags evidence captured significantly away from the registered project location.
43	GIS Project Map	Displays MPLADS projects geographically on an interactive map.
44	India-Level Project Map	MoSPI can view projects across India.
45	State-Level Map	State authorities can view projects within their state.
46	District-Level Map	District authorities can inspect projects in their district.
47	Project Status Indicators	Map can display Green, Orange and Red project indicators.
48	Project Clustering	Allows authorities to identify concentration of projects in particular areas.
49	Duplicate Location Detection	Identifies potentially duplicated/overlapping project locations.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
50	Satellite Image Monitoring	Satellite imagery can provide additional spatial evidence of project-site changes where suitable imagery and resolution are available.
51	Satellite Change Analysis	Compares available imagery over time to identify potential physical changes around a project location.

Important Limitation

Satellite imagery cannot reliably prove every type of construction work. For example, it may be useful for large visible assets but weak for underground work or small structures. Therefore, it should be treated as supporting evidence, not as the sole verification mechanism.

E. Evidence & Image Verification

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
52	In-App Camera	Contractor/authorized field user captures project evidence directly through the application.
53	Controlled Photo Capture	The system encourages evidence to be captured from the authorized application instead of simply uploading old gallery images.
54	Multiple-Angle Photos	The system can require several photographs from different angles for better project assessment.
55	Minimum / Maximum Photo Rules	Project-specific rules can define acceptable evidence quantity.
56	GPS + Timestamp Metadata	Evidence is associated with location and capture time.
57	Duplicate Image Detection	Detects potentially reused photographs across projects or submissions.
58	Image Consistency Analysis	AI can look for inconsistencies between submitted images and project information.
59	AI-Generated / Manipulated Image Screening	AI-based screening can flag potentially manipulated or synthetic images for human review.
60	OCR	Extracts information from uploaded invoices/documents.
61	Document Classification	Identifies whether an uploaded document appears to be an invoice, bill, certificate, etc.
62	Evidence Completeness Check	Checks whether required evidence has been submitted.

Important Note

AI-generated-image detection should produce a risk signal, not a final accusation. Detection technology is not perfect.

F. Financial Monitoring

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
63	Sanction vs Expenditure Comparison	Compares sanctioned amount with actual expenditure.
64	Estimated vs Actual Cost	Identifies significant deviations between estimated and actual cost.
65	Budget Utilization Tracking	Shows how much of the sanctioned amount has been utilized.
66	Component-Wise Expenditure	Breaks expenditure into materials, labour, equipment, etc.
67	Payment Monitoring	Tracks payment patterns against project progress.
68	Cost Overrun Detection	Flags projects where expenditure exceeds expected thresholds.
69	Unusual Expenditure Detection	Detects unusual spending patterns.
70	Expenditure Velocity Analysis	Examines how quickly funds are being spent.
71	Progress vs Expenditure Analysis	Compares financial expenditure with reported physical progress.
72	Historical Cost Benchmarking	Compares similar projects using historical data.
73	Regional Cost Comparison	Compares project costs with relevant regional/project benchmarks.
74	Material Rate Analysis	Checks unusual material prices against available reference/benchmark data.
75	Invoice Amount Analysis	Analyzes invoice amounts for unusual deviations.
76	GST Invoice Verification Support	Checks submitted GST invoice information against available records/data sources where integration is available.

Example

If a water tank is estimated at ₹50 lakh and comparable projects normally fall within a reasonable range around that value, the system does not automatically say: *"This project is fraudulent."* Instead it says: *"Cost anomaly detected — project cost is significantly higher than comparable projects. Review factors such as capacity, material specifications, labour rates, location and design."* That is much more defensible.

G. Duplicate & Irregularity Detection

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
77	Duplicate Project Detection	Identifies potentially duplicated project records.
78	Duplicate Contractor Allocation Detection	Flags cases where the same work appears associated with multiple contractors.
79	Duplicate Location Detection	Finds projects registered at the same/nearby coordinates.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
80	Duplicate Document Detection	Identifies potentially reused documents.
81	Duplicate Image Detection	Finds potentially reused progress photographs.
82	Repeated Invoice Pattern Detection	Identifies suspiciously repeated invoice patterns.
83	Project Overlap Detection	Identifies potentially overlapping works.
84	Entity Relationship Analysis	Examines relationships between projects, agencies and contractors to identify unusual patterns.

H. Project Progress & Delay Intelligence

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
85	Milestone Monitoring	Tracks whether project milestones are being achieved.
86	Expected vs Actual Progress	Compares planned progress with reported progress.
87	Delay Detection	Identifies projects that exceed expected timelines.
88	Delay Severity Analysis	Categorizes delays according to defined thresholds.
89	Progress Stagnation Detection	Flags projects where progress remains unchanged for an unusual period.
90	Progress-Expenditure Mismatch	Flags cases where high expenditure is reported but physical progress appears low.
91	Completion Prediction	Estimates whether a project is likely to complete within its planned timeline.
92	Delay Prediction	Predicts potential future delays based on historical/project data.

I. Compliance Monitoring

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
93	Rule Engine	Stores configurable MPLADS monitoring rules and validation conditions.
94	Norm Compliance Check	Checks submitted project data against applicable requirements.
95	Document Compliance	Checks whether required documents have been submitted.
96	Tender Compliance Support	Identifies missing or inconsistent tender-related information where relevant data is available.
97	Financial Compliance	Checks expenditure/payment information against defined rules.
98	Timeline Compliance	Checks whether projects are progressing within applicable timelines.
99	Evidence Compliance	Checks whether required progress evidence has been submitted.

NO.	SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
100	Automated Compliance Alerts	Generates alerts when defined requirements are not satisfied.

J. AI Anomaly Detection Engine

This is the core intelligence layer.

NO.	ANOMALY TYPE	WHAT THE SYSTEM CHECKS
101	Cost Anomaly	Is the cost unusually high/low compared with appropriate benchmarks?
102	Expenditure Anomaly	Is spending occurring in an unusual pattern?
103	Payment Anomaly	Are payments inconsistent with project status or expected patterns?
104	Invoice Anomaly	Are invoice values or patterns unusual?
105	Material Price Anomaly	Is a reported material rate significantly different from relevant benchmarks?
106	Duplicate Work Anomaly	Does another project appear to represent the same/overlapping work?
107	Location Anomaly	Is the evidence being submitted from an unexpected location?
108	Image Anomaly	Is evidence potentially reused/manipulated/inconsistent?
109	Progress Anomaly	Does reported progress differ significantly from expected progress?
110	Delay Anomaly	Is project execution unusually delayed?
111	Progress-Expenditure Anomaly	Is expenditure inconsistent with physical progress?
112	Compliance Anomaly	Is required information or procedure apparently missing/deviating?
113	Geospatial Anomaly	Does GIS/satellite/project-location data show an unusual pattern?
114	Relationship Anomaly	Are there unusual relationships or repeated patterns among projects/entities?
115	Recurring Pattern Detection	Does a similar anomaly repeatedly occur across projects or entities?

K. Predictive Analytics

SERVICE / FEATURE	WHAT IT ACTUALLY MEANS
Delay Prediction	Estimates the probability that a project may be delayed.
Cost Overrun Prediction	Estimates whether current trends may lead to cost escalation.
Risk Prediction	Identifies projects likely to become high-risk.
Project Outcome Prediction	Predicts likely project execution outcomes using historical/current data.
Trend Forecasting	Identifies emerging expenditure, delay or anomaly trends.

L. Risk Scoring

Instead of presenting hundreds of raw anomalies, the system combines signals.

FEATURE	DESCRIPTION
Risk Score 0–100	Converts multiple risk signals into an understandable score.
Risk Factors	Shows exactly which factors contributed to the score.
Risk Severity	Classifies projects into Normal, Attention or High Risk.
Risk Prioritization	Helps authorities focus on the highest-priority cases first.
Project Risk Profile	Shows the financial, execution, geographic, evidence and compliance risk of a project.

Example

Project Risk Score: 82/100

Reasons: 24% cost deviation, 45-day delay, expenditure higher than physical progress, GPS mismatch, unusual invoice rate.

The authority can then investigate the case.

M. Explainable AI

This is essential.

FEATURE	DESCRIPTION
Why Flagged?	Explains why an anomaly was detected.
Evidence Used	Shows which data contributed to the alert.
Risk Factors	Shows the strongest contributing factors.
Benchmark Comparison	Shows comparison against relevant projects/rules.
Recommended Action	Suggests what the authority should examine next.
Confidence Indicator	Indicates the confidence/strength of the analytical signal.

The system should answer: **What happened?** → **Why was it flagged?** → **What evidence supports it?** → **How serious is it?** → **What should the authority verify?**

N. Recommendation Engine

FEATURE	DESCRIPTION
Corrective Action Suggestions	Suggests possible next steps based on the anomaly.
Evidence Request	Recommends requesting additional documents/photos.

FEATURE	DESCRIPTION
Field Verification Recommendation	Suggests physical verification when required.
Financial Review Recommendation	Suggests financial examination for financial anomalies.
Priority Recommendation	Determines which cases should receive immediate attention.
Monitoring Recommendation	Suggests continued monitoring for lower-risk cases.

Again, recommendation ≠ automatic government action.

O. Smart Alert System

FEATURE	DESCRIPTION
 Green — Normal	No significant anomaly detected. Continue normal monitoring.
 Orange — Attention	Anomaly requires review or additional evidence.
 Red — High Risk	Significant risk indicators; human verification should be prioritized.
Role-Based Routing	Sends the alert to the appropriate authority.
Notification System	Dashboard/app/email notifications can be generated.
Escalation	Unresolved high-priority cases can be escalated according to configured rules.
Alert History	Maintains historical records of alerts.
Case Management	Converts an alert into a trackable verification case.

P. Role-Based Dashboards

1. MoSPI — National Level

MoSPI sees: All-India projects, State-wise performance, MP-wise/project-wise trends, Total expenditure, Project completion, Risk distribution, Anomaly hotspots, India GIS map, High-risk projects, Delay trends, Cost trends, Comparative analytics, Reports.

2. State Authority

Sees: State projects, MPs within the state, District performance, Fund utilization, Risk distribution, Project delays, Anomaly cases, State GIS map, Reports.

3. District Authority (Primary Operational Monitoring Level)

Sees: All projects in district, Recommendations, Sanctions, Implementing agencies, Contractors, Progress, Bills, Invoices, GPS evidence, Risk alerts, Verification cases, Corrective actions.

4. MP

Sees: Recommended works, Sanction status, Project progress, Expenditure, Contractor/agency information, Risk alerts related to their constituency, GIS location, Project evidence, Anomaly explanations.

5. Implementing Agency

Sees: Assigned projects, Tender/contract information, Progress, Contractor information, Required submissions, Project documentation, Alerts relevant to its works.

6. Contractor (Restricted Access)

Sees only assigned projects and permitted actions: Project details, Timeline, Progress submission, Camera, Geo-tagged photos, Bills, GST invoices, Documents, Submission status, Correction requests.

Q. Human Verification & Case Management

This is what prevents the system from becoming dangerously over-automated.

STAGE	DESCRIPTION
1. Alert Generated	AI detects an unusual pattern.
2. Case Created	System creates a review case.
3. Authority Reviews	Authorized official examines the evidence.
4. Additional Evidence Requested	Contractor/agency may be asked to provide additional evidence.
5. Field Verification	Physical inspection can be conducted if necessary.
6. Finding Recorded	Authority records the actual finding.
7. Legitimate / Resolved	Anomaly has a valid explanation or is resolved.
8. Concern Confirmed	Concern requires corrective/administrative action.
9. Corrective Action	Appropriate authority takes action under existing government procedures.
10. Case Closure	Final outcome is recorded.

R. Audit & Transparency

FEATURE	DESCRIPTION
Complete Audit Trail	Records important actions performed within the platform.
User Activity Logs	Records who submitted/modified/reviewed information.
Document Versioning	Maintains versions of important documents.
Evidence History	Maintains evidence submission history.
Alert History	Records generated and resolved alerts.
Case History	Stores complete verification lifecycle.
Decision History	Records actions taken by authorized officials.
Data Provenance	Tracks where analytical data originated.

S. Data Architecture & Integration

FEATURE	DESCRIPTION
Central Data Platform	Provides a unified data layer for MPLADS monitoring information.
PostgreSQL	Stores structured application/project/financial information.
PostGIS	Stores and analyzes geographical information.
Object Storage	Stores photographs, documents and other evidence.
Historical Database	Stores historical project data for benchmarking and ML.
API Integration	Allows integration with relevant government systems where APIs are available.
Data Standardization	Converts data from different sources into common formats.
Data Deduplication	Removes/flags duplicate records.
Entity Resolution	Identifies when different records refer to the same entity.
Data Provenance	Maintains source information for data used by analytics.

2. Complete System Flow

The structured end-to-end operational and analytical lifecycle of MPLADS IntelliTrack is illustrated below:

01

Start & Work Recommendation

MP recommends the work; District Scrutiny & Verification takes place.



02

Sanction & Assignment

Existing Government Process approval, Implementing Agency assignment, Tender, and Contractor selection with Unique Project ID generation.



03

Data Collection & Validation

Sanction, Cost, Expenditure, Payments, Progress, Bills, GST Invoices, Documents, GPS Photos, GIS, Satellite, and Contractor Data are collected and validated (Schema, Completeness, Consistency, Duplicates, Location, Evidence). Valid data flows forward; Invalid data triggers correction requests.



04

Unified Data Store

Centralized storage combining Project, Financial, GIS, Evidence, Documents, and Historical data.



05

Intelligence & Analysis Engines

AI/ML Engine (Financial, Execution, Geospatial, Evidence, Compliance, Trend, Relationship), Compliance & Rule Engine, Anomaly Detection Engine (Cost, Expenditure, Payment, Invoice, Duplicate, Location, Image, Progress, Delay), and Predictive Engine.



06

Risk Scoring & Explainable AI

Aggregate signals into a 0–100 Risk Score and Classification. Explainable AI answers: What happened? Why flagged? What evidence? Severity? What to verify?



07

Recommendations & Smart Alert Routing

Recommendation Engine suggests actions. Smart Alert System routes alerts via Green (Normal), Orange (Attention), and Red (High Risk) indicators.



08

Role-Based Dashboards & Human Verification

MoSPI, State, District, MP, Agency, and Contractor dashboards. Human verification workflow: Review evidence, request more evidence, or conduct field verification.



09

Outcome, Audit Trail & Model Feedback

Verified/Legitimate (continue monitoring) vs Concern Confirmed (corrective action). All actions recorded in Audit Trail, feeding into Model Feedback & Continuous Learning loop.

3. The Simplified Logic

CORE PLATFORM LOGIC (12 WORDS)

**Recommend → Scrutinize → Sanction → Execute → Collect → Validate → Analyze →
Detect → Predict → Alert → Verify → Improve**

That is the core of MPLADS IntelliTrack.

What makes your solution different

The system is not simply:

“AI detects fraud.”

That would be weak and unrealistic.

Your stronger proposition is:

“The system continuously analyzes MPLADS project, financial, execution, geospatial and evidence data to identify anomalies, assess risk, explain the reasons behind the risk, provide actionable recommendations, and support authorities in timely human verification and corrective action.”

That wording matches the actual SIH problem much more closely because the official problem asks for anomaly detection, fraud-risk identification, inefficiency detection, trend analysis, predictive insights, risk-based alerts, compliance monitoring and decision support—while your architecture adds the operational evidence, GIS, satellite, explainability, human verification and feedback layers around those requirements.